

Enterprise VLAN & Trunking Lab

Phase 4 Report – IEEE 802.1Q Trunk Configuration

Project Information

Project Name: Enterprise VLAN & Trunking Lab

Phase: Phase 4 – IEEE 802.1Q Trunk Configuration

Simulation Software: Cisco Packet Tracer

Devices Configured: SW1, SW2

Status: Completed

1. Introduction

Following the successful implementation of VLANs and access ports in Phase 3, the next step in the deployment of the enterprise network was to establish communication between switches through an IEEE 802.1Q trunk link.

A trunk link enables multiple VLANs to traverse a single physical connection by tagging Ethernet frames with VLAN information. This mechanism allows switches to preserve VLAN membership across the network while maintaining logical separation between broadcast domains.

During this phase, the GigabitEthernet0/1 interfaces on both switches were configured as trunk ports, a dedicated native VLAN was assigned, and the trunk was restricted to carrying only the VLANs required by the network design.

2. Objectives

The objectives of this phase were to:

- Configure an IEEE 802.1Q trunk between SW1 and SW2.
- Convert the inter-switch uplink from an access port to a trunk port.
- Configure a dedicated native VLAN.
- Restrict the trunk to carry only authorized VLANs.
- Verify trunk operation using Cisco IOS commands.
- Prepare the enterprise network for VLAN communication across multiple switches.

3. Why Trunking is Required

In an enterprise environment, multiple VLANs often span more than one switch. Without a trunk link, each VLAN would require a separate physical cable between switches, resulting in unnecessary complexity and poor scalability.

IEEE 802.1Q solves this problem by inserting a VLAN tag into Ethernet frames, allowing multiple VLANs to share the same physical connection while maintaining logical separation.

The trunk configured during this phase enables future expansion of the network without requiring additional inter-switch links.

4. Trunk Design

The trunk link was established between the following interfaces:

Device	Interface
SW1	GigabitEthernet0/1
SW2	GigabitEthernet0/1

This connection serves as the backbone between the two access switches.

5. Native VLAN Configuration

A dedicated native VLAN was configured for the trunk.

Parameter	Value
Native VLAN	99

Frames belonging to VLAN 99 are transmitted untagged across the trunk, while frames from all other VLANs are tagged using the IEEE 802.1Q standard.

Using a dedicated native VLAN instead of the default VLAN 1 is considered an enterprise networking best practice.

6. Allowed VLAN Configuration

To improve security and reduce unnecessary traffic, the trunk was configured to carry only the VLANs used within the network.

The following VLANs were permitted:

VLAN ID	Department
10	Management
20	Human Resources
30	Sales
40	Information Technology
99	Native VLAN

Restricting the allowed VLAN list prevents unused VLANs from traversing the trunk and reduces the attack surface of the network.

7. Trunk Configuration Summary

The following configuration elements were applied to both switches:

- Interface converted to trunk mode.
- Native VLAN configured as VLAN 99.
- Allowed VLAN list configured.
- Interface description added for documentation.
- Configuration saved to startup memory.

Both SW1 and SW2 received identical trunk configurations to ensure proper operation.

8. Verification

The following Cisco IOS commands were used to verify the trunk configuration.

Trunk Status Verification

The command:

```
show interfaces trunk
```

confirmed:

- GigabitEthernet0/1 operating in trunk mode.
 - IEEE 802.1Q encapsulation in use.
 - Native VLAN configured as VLAN 99.
 - Allowed VLAN list correctly configured.
 - Active VLANs matching the network design.
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Interface Verification

The following command was executed:

```
show interfaces gigabitEthernet0/1 switchport
```

Verification confirmed:

- Administrative Mode: Trunk
 - Operational Mode: Trunk
 - Native VLAN: 99
 - Allowed VLAN List: 10,20,30,40,99
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Running Configuration Verification

The following command was used:

```
show running-config interface gigabitEthernet0/1
```

The configuration review confirmed:

- Trunk mode enabled.
 - Native VLAN correctly assigned.
 - Allowed VLAN list configured.
 - Interface description present.
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VLAN Verification

The command:

```
show vlan brief
```

confirmed that GigabitEthernet0/1 was no longer listed under VLAN 1.

This behavior is expected because trunk ports are not members of a single access VLAN; instead, they transport traffic for multiple VLANs simultaneously.

9. Connectivity Considerations

The current laboratory topology assigns each department to a single switch:

SW1

- Management
- Human Resources

SW2

- Sales
- Information Technology

Since no VLAN currently spans both switches, the trunk configuration does not alter the results of the existing connectivity tests. Communication within each VLAN continues to occur locally on the same switch.

However, the trunk establishes the infrastructure required for future scenarios in which the same VLAN is distributed across multiple switches.

10. Technical Concepts Applied

This phase reinforced several important Layer 2 networking concepts:

- IEEE 802.1Q Trunking
- VLAN Tagging
- Native VLAN
- Allowed VLAN Lists
- Inter-switch Communication
- Enterprise Switch Uplinks
- Cisco IOS Trunk Configuration
- Layer 2 Network Scalability

These concepts are fundamental to enterprise campus network design and are widely used in production environments.

11. Best Practices Applied

The following best practices were implemented during this phase:

- Configured a dedicated trunk interface.
- Used a non-default native VLAN.
- Restricted the trunk to required VLANs only.
- Added descriptive interface labels.
- Verified the configuration after implementation.
- Saved the running configuration to startup memory.

These practices contribute to improved network security, maintainability, and operational consistency.

12. Deliverables

The following deliverables were completed during Phase 4:

- IEEE 802.1Q trunk configured on SW1.
 - IEEE 802.1Q trunk configured on SW2.
 - Native VLAN configured.
 - Allowed VLAN list implemented.
 - Trunk verified using Cisco IOS commands.
 - Interface configuration documented.
 - Configuration saved to startup memory.
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13. Skills Developed

By completing this phase, the following practical skills were developed:

- Configuring trunk interfaces
- Understanding IEEE 802.1Q encapsulation
- Configuring native VLANs
- Restricting allowed VLANs
- Verifying trunk operation
- Troubleshooting Layer 2 switch uplinks
- Applying enterprise switch configuration standards

These competencies are essential for Network Engineers responsible for campus switching infrastructures.

14. Lessons Learned

Several key networking principles were demonstrated during this phase:

- Trunk ports allow multiple VLANs to share a single physical connection.
- IEEE 802.1Q uses VLAN tagging to preserve VLAN membership between switches.
- Native VLAN traffic is transmitted without an 802.1Q tag.
- Restricting allowed VLANs improves security and efficiency.
- Trunk ports are not displayed as members of individual VLANs in the VLAN database because they transport traffic for multiple VLANs.

Understanding these concepts is critical for designing scalable and secure enterprise LANs.

15. Preparation for Phase 5

With the VLAN infrastructure and trunk link successfully configured, the enterprise switching environment is now operational.

The next phase will focus on:

- End-to-end verification of the Layer 2 network.
 - Connectivity testing.
 - MAC address table analysis.
 - Switch forwarding behavior.
 - Troubleshooting common VLAN and trunk issues.
 - Validation of the complete enterprise LAN deployment.
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16. Conclusion

Phase 4 successfully established an IEEE 802.1Q trunk between the two Cisco switches, enabling the transport of multiple VLANs over a single physical link. The trunk was configured according to enterprise best practices by assigning a dedicated native VLAN, restricting the allowed VLAN list, and verifying proper operation using Cisco IOS diagnostic commands.

This phase completed the Layer 2 infrastructure required for a scalable enterprise switched network. With the trunk in place, the network is prepared for future expansion, inter-switch VLAN communication, and more advanced switching technologies introduced in subsequent projects.